

Prediction and Effect Size Estimation

A prediction problem begins with the observation of n independent vectors, the “training set”,

$$(\mathbf{x}_j, y_j) \quad j = 1, 2, \dots, n, \quad (11.1)$$

where $\mathbf{x}_j$ is an N -vector of predictors and y_j a real-valued response. Using the training set, the goal is to construct an effective prediction rule $r(\mathbf{x})$: having observed a new vector $\mathbf{x}$ but not y , it is hoped that $r(\mathbf{x})$ will accurately predict y . An insurance company, for instance, might collect predictors (age, gender, smoking habits) and be interested in predicting imminent heart attacks.

Classical prediction methods depend on Fisher’s linear discriminant function. Here the response variable is dichotomous, $y = 1$ or 2 (perhaps representing “healthy” or “sick”) and $\mathbf{x}_j$ has a multivariate normal distribution whose mean vector depends on y ,

$$y = \begin{cases} 1 & \mathbf{x} \sim \mathcal{N}_N(\boldsymbol{\delta}_1, \boldsymbol{\Sigma}) \\ 2 & \mathbf{x} \sim \mathcal{N}_N(\boldsymbol{\delta}_2, \boldsymbol{\Sigma}). \end{cases} \quad (11.2)$$

In the Bayesian situation, where y has prior probabilities¹

$$p_1 = \Pr\{y = 1\} \quad \text{and} \quad p_2 = \Pr\{y = 2\} \quad (11.3)$$

an application of Bayes rule gives log posterior odds ratio

$$\log \left(\frac{\Pr\{y = 2|\mathbf{x}\}}{\Pr\{y = 1|\mathbf{x}\}} \right) = \beta_0 + \boldsymbol{\beta}'\mathbf{x} \quad (11.4)$$

with

$$\begin{aligned} \boldsymbol{\beta}' &= (\boldsymbol{\delta}_2 - \boldsymbol{\delta}_1)' \boldsymbol{\Sigma}^{-1} \\ \beta_0 &= \log(p_2/p_1) + (\boldsymbol{\delta}_2' \boldsymbol{\Sigma}^{-1} \boldsymbol{\delta}_2 - \boldsymbol{\delta}_1' \boldsymbol{\Sigma}^{-1} \boldsymbol{\delta}_1)/2. \end{aligned} \quad (11.5)$$

¹ Notice that this is different than the two-groups model (2.7), where the distinction is between the unobserved null vs non-null dichotomy, rather than healthy vs sick, which is observable in the training set.

Exercise 11.1 Verify (11.4)–(11.5).

The optimum Bayes rule predicts $y = 1$ or 2 as $\beta_0 + \beta'x$, *Fisher's linear discriminant function*, is less than or greater than 0 ; (11.4) is a nice linear function of x , neatly dividing the space of possible x vectors into prediction regions separated by the hyperplane $\beta_0 + \beta'x = 0$. In practice, β_0 and β are estimated from the training set using the usual unbiased estimates of δ_1 , δ_2 , and Σ in model (11.2), with $\log(p_2/p_1)$ perhaps estimated by $\log(n_2/n_1)$, the observed numbers in the two categories.

All of this works well in the classic situation where the number of independent observations n is much bigger than N , the predictor dimension. This is just what we *don't* have in large-scale applications. Suppose we wish to use the prostate data set of Section 2.1 to develop a prediction rule for prostate cancer: given x , a microarray expression vector for the $N = 6033$ genes, we want a rule $r(x)$ that predicts whether the man who provided x will develop prostate cancer. Now $N = 6033$, $n = 102$, and the usual unbiased estimates of (11.4) will probably be useless, unbiasedness being an unaffordable luxury in high dimensions.

A variety of computer-intensive prediction algorithms has been developed recently that trade bias for stability and accuracy: boosting, bagging, support vector machines, LARS, Lasso, and ridge regression being some of the prominent candidates. This chapter will discuss a comparatively simple empirical Bayes approach to large-scale prediction. Besides adding to the store of useful methodology, this will give us a chance to put prediction problems into a false discovery rate context.

In the prostate example it seems obvious that a good prediction rule should depend on genes that express themselves much differently in sick and healthy subjects, perhaps those falling beyond the small triangles in Figure 5.2. Fdr analysis aims to identify such genes. In Section 11.4 we will go further, and try to estimate *effect size*, a quantitative assessment of just how non-null a gene may be.

We begin in Section 11.1 with a simple model that leads to an empirical Bayes algorithm for prediction and, eventually, to effect size estimation. We restrict attention to the *dichotomous case* where each y_j equals 1 or 2 , calling the two states “healthy” and “sick” for the sake of definiteness, and refer to the components of x_j as “gene expression” or just “genes”, though of course the theory has nothing particularly to do with genomics.

11.1 A Simple Model

Motivation for our empirical Bayes prediction rules starts from a drastically idealized model for the predictor vectors $\mathbf{x}_j$ in (11.1). We assume that the components x_i of a typical such vector $\mathbf{x} = (x_1, x_2, \dots, x_i, \dots, x_N)'$ are independently normal, with location and scale parameters μ_i and σ_i , and with possibly different expectations in the two subject categories,

$$\frac{x_i - \mu_i}{\sigma_i} \underset{\text{ind}}{\sim} \mathcal{N}\left(\pm \frac{\delta_i}{2c_0}, 1\right) \quad \begin{cases} \text{"-"} & \text{healthy class (y = 1)} \\ \text{"+"} & \text{sick class (y = 2)}. \end{cases} \quad (11.6)$$

The constant c_0 equals

$$c_0 = (n_1 n_2 / n)^{1/2} \quad (11.7)$$

where n_1 and n_2 are the number of healthy and sick subjects in the training set, $n = n_1 + n_2$. (Dividing by c_0 in (11.6) will make δ_i the effect size in what follows.) Null genes have $\delta_i = 0$, indicating no difference between the two categories; non-null cases, particularly those with large values of $|\delta_i|$, are promising ingredients for effective prediction.

Let

$$u_i = (x_i - \mu_i) / \sigma_i \quad i = 1, 2, \dots, N, \quad (11.8)$$

be the standardized version of x_i in (11.7) so that $\mathbf{u}$ has two possible N -dimensional normal distributions,

$$\mathbf{u} \sim \mathcal{N}_N(\pm \boldsymbol{\delta} / (2c_0), \mathbf{I}), \quad (11.9)$$

"-" or "+" as $y = 1$ or 2 , with $\boldsymbol{\delta} = (\delta_1, \delta_2, \dots, \delta_N)'$ the vector of effect sizes and $\mathbf{I}$ the $N \times N$ identity matrix.

The ideal prediction rule depends on the weighted sum

$$S = \sum_{i=1}^N \delta_i u_i \sim \mathcal{N}\left(\pm \|\boldsymbol{\delta}\|^2 / 2c_0, \|\boldsymbol{\delta}\|^2\right) \quad (11.10)$$

with "-" and "+" applying as in (11.6). If the two categories in (11.3) have equal prior probabilities,

$$p_1 = p_2 = 0.50 \quad (11.11)$$

we predict

$$\begin{aligned} &\text{healthy (y = 1) if } S < 0 \\ &\text{sick (y = 2) if } S > 0. \end{aligned} \quad (11.12)$$

Prediction error rates of the first and second kinds, i.e., confusing healthy with sick or vice versa, both equal

$$\alpha \equiv \Phi(-\|\delta\|/2c_0). \quad (11.13)$$

We need $\|\delta\| = (\sum \delta_i^2)^{1/2}$, the length of the effect size vector, to be large for successful prediction.

Exercise 11.2 Use Fisher's linear discriminant function to verify (11.10)–(11.13).

The ideal rule (11.12) is unavailable even if we believe model (11.6): in practice we need to estimate the parameters

$$(\mu_i, \sigma_i, \delta_i) \quad i = 1, 2, \dots, N, \quad (11.14)$$

entering into S (11.10), more than 18 000 of them for the prostate data. This is where the training data (11.1) comes in,

$$X_{N \times n} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n) \quad \text{and} \quad \mathbf{y} = (y_1, y_2, \dots, y_n)'. \quad (11.15)$$

By reordering the subjects, we can take the first n_1 entries $y_j = 1$ and the last n_2 entries $y_j = 2$, that is, healthy subjects first. Let $\bar{x}_{i1}$ and SS_{i1} be the mean and within-group sum of squares for gene i measurements in the healthy subjects,

$$\bar{x}_{i1} = \sum_{j=1}^{n_1} x_{ij}/n_1 \quad \text{and} \quad SS_{i1} = \sum_{j=1}^{n_1} (x_{ij} - \bar{x}_{i1})^2 \quad (11.16)$$

and similarly $\bar{x}_{i2}$ and SS_{i2} for the n_2 sick subjects. Then

$$\hat{\mu}_i = \frac{\bar{x}_{i1} + \bar{x}_{i2}}{2} \quad \text{and} \quad \hat{\sigma}_i^2 = \frac{SS_{i1} + SS_{i2}}{n - 2} \quad (11.17)$$

are unbiased estimates of μ_i and σ_i^2 .

The crucial prediction parameters are the effect sizes δ_i . This is where unbiasedness fails us. Assume for the moment that σ_i is known, in which case

$$\bar{\delta}_i \equiv c_0 \frac{\bar{x}_{i2} - \bar{x}_{i1}}{\sigma_i} \sim \mathcal{N}(\delta_i, 1) \quad (11.18)$$

unbiasedly estimates δ_i . Substituting $\hat{\sigma}_i$ for σ_i gives the two-sample t -statistic t_i (2.2) that we can transform to z_i as in (2.5) with approximate distribution

$$z_i \sim \mathcal{N}(\delta_i, 1), \quad (11.19)$$

which becomes increasingly accurate for large values of n . For now we

will treat it as exact, though the prediction program Ebay introduced in Section 11.2 makes a small change to accommodate the effects of transformation (2.5).

This looks promising. For prediction purposes, however, we are hoping to find extreme values of δ_i , either positive or negative, while *selection bias* makes the extreme z_i values overinflated estimates: $z_{610} = 5.29$ is the most extreme z -value for the prostate data, but it is likely that δ_{610} is less than 5.29; z_{610} has won a “farthest from the origin” contest among $N = 6033$ contenders, partly no doubt from having δ_{610} large, but also from having a positive random error (or else it probably would not have won the contest), which is what selection bias means.

The **pamr** algorithm² counteracts selection bias by shrinking the estimates z_i toward zero according to what is called a *soft thresholding rule*,

$$\hat{\delta}_i = \text{sign}(z_i) \cdot (|z_i| - \lambda)_+ \quad (11.20)$$

where x_+ equals $\max(x, 0)$. That is, each z_i is shrunk toward zero by some fixed amount λ , under the restriction that shrinking never goes past zero. A range of possible shrinkage parameters λ is tried, and for each one a statistic like (11.10) is formed from the estimates $(\hat{\mu}_i, \hat{\sigma}_i, \hat{\delta}_i)$,

$$\hat{S}_\lambda(\mathbf{x}) = \sum_{i=1}^N \hat{\delta}_i \hat{u}_i \quad [\hat{u}_i = (x_i - \hat{\mu}_i) / \hat{\sigma}_i] \quad (11.21)$$

leading to the rule $r_\lambda(\mathbf{x})$ that predicts $y = 1$ or 2 as $\hat{S}_\lambda(\mathbf{x})$ is less than or greater than zero (11.12). A cross-validation method, described below, is then employed to estimate α_λ , the error rate for $r_\lambda(\mathbf{x})$ (as well as providing separate estimates for errors of the first and second kinds).

Table 11.1 shows **pamr** output for the prostate data. As λ increased from 0, the estimated error rate $\hat{\alpha}_\lambda$ decreased, reaching a minimum of 0.08 at $\lambda = 2.16$ and then increasing again. This suggests that rule $r_{2.16}(\mathbf{x})$ would make only 8% errors in predicting prostate cancer from microarray measurements.

Notice that the prediction statistic $\hat{S}_\lambda(\mathbf{x})$ involves only those genes with $|z_i| > \lambda$, since $\hat{\delta}_i$ (11.20) is zero for $|z_i| \leq \lambda$. At $\lambda = 2.16$, only 377 of the 6033 genes are involved in the prediction. Less is more as far as prediction is concerned, rules based on fewer predictions being generally easier to implement and more reproducible.

² *Prediction analysis for microarrays*, also known as the “nearest shrunken centroid” algorithm, is available from the R library CRAN.

Table 11.1 Cross-validated error rates $\hat{\alpha}_\lambda$, **pamr** prediction rules applied to prostate cancer data, for increasing choices of shrinkage parameter λ (11.20)–(11.21). Taking $\lambda = 2.16$ gave $\hat{\alpha}_\lambda = 0.08$, using a prediction rule based on 377 genes.

Shrinkage λ	# Genes	$\hat{\alpha}_\lambda$
.00	6033	.34
.54	3763	.33
1.08	1931	.23
1.62	866	.12
2.16	377	.08
2.70	172	.10
3.24	80	.16
3.78	35	.30
4.32	4	.41
4.86	1	.48
5.29	0	.52

Cross-validation

Ten-fold cross-validation, used in **pamr**, randomly splits the n subjects into ten “folds” with roughly proportional numbers of “healthy” and “sick” in each fold. The prediction algorithm is refit ten times with the cases of each fold withheld from the training set in turn, the cross-validated error rate $\hat{\alpha}_{CV}$ being the average error rate on the withheld cases. A typical fold contained ten subjects for the prostate data, five healthy and five sick, who were then predicted by the rule constructed from the data of the other 92 subjects; **pamr** actually repeats ten-fold cross-validation several times, averaging the cross-validated error rates over all the repetitions.

Apparent error, how accurately a rule predicts its own training set, is usually too optimistic. Cross-validation is a simple way to estimate, almost unbiasedly, how well the rule will do on a new set of data. It is important to remember that $\hat{\alpha}_{CV}$ is *not* an estimate of error for a specific rule, for instance $\hat{r}_{2.16}(x)$ in Table 11.1. Rather, it is the expected error rate for rules selected according to the same recipe. Cross-validation is not perfect and can give misleadingly optimistic estimates in situations like (6.36), but it is almost always preferable to the apparent error rate.

Examining Table 11.1, it seems we should use $\lambda = 2.16$ for our prediction rule. However, there is a subtle danger lurking here: because we

have looked at all the data to select the “best” λ , this choice is not itself cross-validated, and the corresponding rate 0.08 may be optimistic.

A small simulation study was run with $N = 1000$, $n_1 = n_2 = 10$, and all $x_{ij} \stackrel{\text{ind}}{\sim} \mathcal{N}(0, 1)$. In this situation, all δ_i in (11.6) are zero and $\alpha = 0.50$ at (11.13); but the minimum cross-validated pamr error rates in 100 simulations of this set-up had median 0.30 with standard deviation ± 0.16 . This is an extreme example. Usually the over-optimism is less severe, particularly when good prediction is possible.

11.2 Bayes and Empirical Bayes Prediction Rules

We now assume a Bayes prior distribution $g(\delta)$ for the effect size parameters δ_i in model (11.6). Reducing the data to z -values (11.19), we have, as in (2.47),

$$\delta \sim g(\cdot) \quad \text{and} \quad z|\delta \sim \mathcal{N}(\delta, 1) \quad (11.22)$$

for a typical (δ, z) pair. For convenience we will treat $g(\delta)$ as a density function but allow it to have discrete atoms, perhaps with an atom π_0 at $\delta = 0$ as in the two-groups model (5.1), though that is not assumed here.

The *normal hierarchical model* (11.22) is a favorite in the empirical Bayes literature because it leads to particularly simple Bayesian estimates.

Theorem 11.1 *Let $f(z)$ be the marginal density of z in model (11.22),*

$$f(z) = \int_{-\infty}^{\infty} \varphi(z - \delta)g(\delta)d\delta \quad \left[\varphi(z) = e^{-z^2/2} / \sqrt{2\pi} \right]. \quad (11.23)$$

Then the posterior density of δ given z is

$$g(\delta|z) = e^{z\delta - \psi(z)} \left(e^{-\delta^2/2} g(\delta) \right) \quad (11.24)$$

where

$$\psi(z) = \log(f(z)/\varphi(z)). \quad (11.25)$$

Proof According to Bayes theorem,

$$g(\delta|z) = \varphi(z - \delta)g(\delta)/f(z). \quad (11.26)$$

□

Exercise 11.3 Complete the proof.

Formula (11.24) is an exponential family of densities for δ with canonical, or natural, parameter z . The standard exponential family properties reviewed in Appendix A then yield the following result.

Corollary 11.2 Under model (11.22), δ has posterior mean and variance

$$E\{\delta|z\} = \psi'(z) \quad \text{and} \quad \text{var}\{\delta|z\} = \psi''(z) \quad (11.27)$$

where ψ' and ψ'' indicate the first and second derivatives of $\psi(z)$. Going further, the k th cumulant of δ given z equals $d^k\psi(z)/dz^k$. Letting $l(z) = \log(f(z))$, we can rewrite (11.27) as

$$E\{\delta|z\} = z + l'(z) \quad \text{and} \quad \text{var}\{\delta|z\} = 1 + l''(z), \quad (11.28)$$

now with primes indicating derivatives of $l(z)$.

The great advantage of Corollary 11.2 is that the moments of δ given z are obtained directly from the marginal density of $f(z)$ without requiring explicit calculation of the prior $g(\delta)$, thus avoiding the usual difficulties of deconvolution. This is essential for empirical Bayes applications, where now we need only estimate $f(z)$, not $g(\delta)$. The R algorithm³ Ebay uses Poisson regression on histogram counts (5.11)–(5.15) to obtain a smooth estimate $\hat{l}(z)$ of $l(z) = \log(f(z))$, and then estimates of $E\{\delta|z\}$ and $\text{var}\{\delta|z\}$,

$$z \longrightarrow \hat{l}(z) \longrightarrow \hat{l}'(z) \longrightarrow \hat{E}\{\delta|z\} = z + \hat{l}'(z) \quad (11.29)$$

and $\widehat{\text{var}}\{\delta|z\} = 1 + \hat{l}''(z)$.

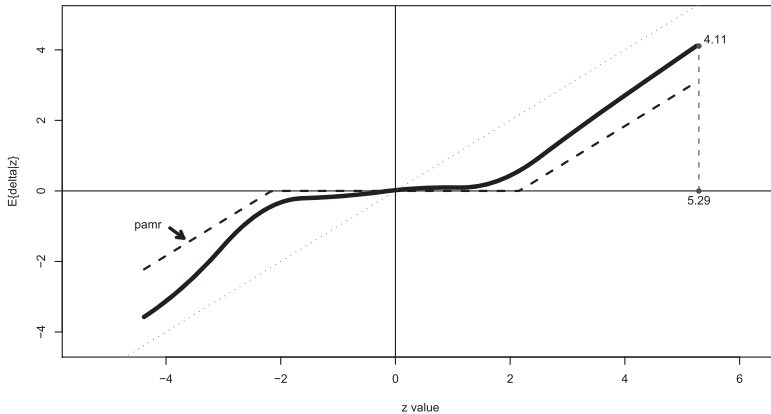


Figure 11.1 Estimated effect size $\hat{E}\{\delta|z\}$, from application of Ebay algorithm to prostate cancer z -values. Dashed curve is soft thresholding rule (11.20) using `pamr` value $\lambda = 2.16$.

The heavy curve in Figure 11.1 shows $\hat{E}\{\delta|z\}$ obtained by application

³ See Appendix B.

of Ebay to the prostate cancer z -values. We see that $\hat{E}\{\delta|z\}$ is nearly zero for z in $(-2, 2)$, then grows almost linearly for $|z| > 2$. Gene 610, with $z_{610} = 5.29$, is estimated to have effect size 4.11. $\hat{E}\{\delta|z\}$ resembles a soft thresholding rule (11.20), but with λ noticeably smaller than the *pamr* choice 2.16, the latter being a more vigorous shrinker, particularly for negative values of z .

Beginning with a vector $\mathbf{z}$ of z -values obtained from t -statistics as in (2.5), the Ebay algorithm produces prediction rules in the following steps:

- 1 A target error rate α_0 is selected (default $\alpha_0 = 0.025$).
- 2 Poisson regression is used to generate effect size estimates

$$\hat{\delta}_i = \hat{E}\{\delta|z_i\} \quad (11.30)$$

as at (11.29), with an adjustment to account for Student- t effects described below at (11.33)–(11.34).

- 3 Letting $\hat{\delta}_m$ be the vector of m largest $\hat{\delta}_i$ values in absolute value, m is selected to be the smallest integer such that the nominal error rate $\hat{\alpha}_m = \Phi(-\|\hat{\delta}_m\|/2c_0)$ (11.13) is less than α_0 , i.e., the minimum m making

$$\|\hat{\delta}_m\| \geq 2c_0\Phi^{-1}(1 - \alpha_0). \quad (11.31)$$

- 4 The resulting empirical Bayes prediction rule is based on the sign of

$$\hat{S} = \sum_m \hat{\delta}_i \hat{u}_i \quad (11.32)$$

as in (11.12), (11.21), the summation being over the m selected cases.

- 5 Repeated ten-fold cross-validation is used to provide an unbiased estimate of the rule's prediction error, as described near the end of Section 11.1.

Table 11.2 displays Ebay's output for the prostate data. At the first step, gene 610 with $z_{610} = 5.29$ and $\hat{\delta}_{610} = 4.11$ was selected, giving $\hat{\alpha} = \Phi(-4.11/2c_0) = 0.342$ ($c_0 = 5.05$ (11.7)). The algorithm stopped at step 55, at which point $\hat{\alpha}_m$ first went below the target value $\alpha_0 = 0.025$.

Cross-validation estimates of prediction error were based on 20 independent replications of ten-fold CV, $\hat{\alpha}_{CV} = 0.090$ being the overall average error (with standard error 0.003, obtained in the usual way for an average). Averaging the CV prediction errors separately over the 50 healthy and 52 sick subjects indicated much poorer accuracy in predicting the latter. Note that here $\hat{\alpha}_{CV}$ is honest in the sense that the form of the empirical Bayes rule is completely determined, as opposed to our *pamr* example where the best value $\lambda = 2.16$ was selected *after* examining the cross-validation results.

Table 11.2 Ebay output for prostate cancer prediction rule (11.32). The rule employs data from the 55 genes having the largest values of $|\hat{\delta}_i|$, the effect size estimates. Also shown is variance estimates $\widehat{\text{var}}$ from (11.27), theoretical prediction error estimates $\hat{\alpha}$, and $\hat{\alpha}_{\text{cor}}$, error estimates including correlations between the genes (11.38).

Step	Index	z	$\hat{\delta}$	$\widehat{\text{var}}$	$\hat{\alpha}$	$\hat{\alpha}_{\text{cor}}$
1	610	5.29	4.11	.87	.342	.342
2	1720	4.83	3.65	.89	.293	.289
3	364	-4.42	-3.57	.92	.258	.258
4	3940	-4.33	-3.52	.92	.231	.231
5	4546	-4.29	-3.47	.93	.208	.223
6	4331	-4.14	-3.30	.97	.190	.197
7	332	4.47	3.24	.91	.175	.189
8	914	4.40	3.16	.92	.162	.175
⋮	⋮	⋮	⋮	⋮	⋮	⋮
45	4154	-3.38	-2.23	1.18	.032	.054
46	2	3.57	2.22	.97	.031	.055
47	2370	3.56	2.20	.97	.031	.053
48	3282	3.56	2.20	.97	.030	.052
49	3505	-3.33	-2.15	1.19	.029	.050
50	905	3.51	2.15	.97	.028	.051
51	4040	-3.33	-2.14	1.19	.027	.052
52	3269	-3.32	-2.12	1.19	.027	.050
53	805	-3.32	-2.12	1.19	.026	.048
54	4552	3.47	2.09	.97	.025	.048
55	721	3.46	2.09	.97	.025	.044

There are many reasons why $\hat{\alpha}_{\text{CV}}$ might exceed the ideal theoretical rate α_0 : $(\hat{\mu}_i, \hat{\sigma}_i)$ (11.17) does not equal (μ_i, σ_i) ; the measurements x_i in (11.6) are not normally distributed; they are not independent; the empirical Bayes estimates $\hat{\delta}_i$ differ from the true Bayes estimates $\delta_i = E\{\delta|z_i\}$ (11.27).

This last point causes trouble at the extreme ends of the z scale, where $|\hat{\delta}(z)|$ is largest but where we have the least amount of data for estimating $E\{\delta|z\}$. An option in Ebay allows for truncation of the $\hat{\delta}$ estimation procedure at some number k_{trunc} of observations in from the extremes. With $k_{\text{trunc}} = 10$, used in the second row of Table 11.3, $\hat{\delta}_i$ for the ten largest z_i values is set equal to $\max\{\hat{\delta}_i : i \leq N - 10\}$, and similarly at the negative end of the z scale. This was moderately effective in reducing the cross-validated prediction errors in Table 11.3.

Table 11.3 Cross-validation estimates of prediction error for prostate cancer data (standard errors in parentheses). Prediction errors are much larger in the sick category. Truncating the prediction rule reduces prediction errors.

	$\hat{\alpha}_{CV}$	Healthy	Sick
no truncation	.090 (.003)	.044 (.005)	.128 (.005)
$k_{\text{trunc}} = 10$	.073 (.004)	.032 (.005)	.107 (.006)

Student- t effects

Section 7.4 showed that the distribution of $z_i = \Phi^{-1}(F_\nu(t_i))$ of a non-central t -statistic with ν degrees of freedom is well-approximated by a modified version of (11.22),

$$z \sim \mathcal{N}(\delta, \sigma^2(\delta)) \quad \left[\sigma^2(\delta) \leq 1 \right] \quad (11.33)$$

where the function $\sigma^2(\delta)$ depends on ν (see Figure 7.6). Rather than formulas (11.29), Ebay uses

$$\hat{E}\{\delta|z\} = z + \sigma^2(z)\hat{l}'(z) \quad \text{and} \quad \widehat{\text{var}}\{\delta|z\} = \sigma^2(z) \left[1 + \sigma^2(z)\hat{l}''(z) \right]. \quad (11.34)$$

This produces slightly larger $\hat{\delta}_i$ estimates since $\sigma^2(z_i) \leq 1$.

Exercise 11.4 (a) Suppose $z \sim \mathcal{N}(\delta, \sigma_0^2)$ for some fixed variance σ_0^2 . Show that

$$E\{\delta|z\} = z + \sigma_0^2 l'(z) \quad \text{and} \quad \text{var}\{\delta|z\} = \sigma_0^2 \left[1 + \sigma_0^2 l''(z) \right]. \quad (11.35)$$

(b) Suggest an improvement on (11.34).

Correlation corrections

The assumption of gene-wise independence in our basic model (11.6) is likely to be untrue, perhaps spectacularly untrue, in applications. It is not difficult to make corrections for correlation structure. Suppose that the vector $\mathbf{u}$ of standardized predictors in (11.8) has covariance matrix Σ rather than $\mathbf{I}$, but we continue to use prediction rule (11.10), (11.12). Then the error probability α (11.13) becomes

$$\alpha = \Phi(-\Delta_0 \cdot \gamma) \quad \text{where} \quad \begin{cases} \Delta_0 = \|\delta\|/2c_0 \\ \gamma = (\delta'\delta/\delta'\Sigma\delta)^{1/2}. \end{cases} \quad (11.36)$$

Here Δ_0 is the independence value while γ is a correction factor, usually less than 1, that increases the error rate α .

Exercise 11.5 Verify (11.36).

The standardized variates u_i have variance 1 so Σ is $\mathbf{u}$'s correlation matrix. At the m th step of the Ebay algorithm we can compute the obvious estimate of γ ,

$$\hat{\gamma}_m = \left(\hat{\delta}_m' \hat{\delta}_m / \hat{\delta}_m' \hat{\Sigma}_m \hat{\delta}_m \right)^{\frac{1}{2}} \tag{11.37}$$

where $\hat{\Sigma}_m$ is the $m \times m$ sample correlation matrix for the first-selected m genes. The last column of Table 11.2 gives the correlation-corrected error estimate

$$\hat{\alpha}_{\text{cor}} = \Phi \left(-\frac{\|\hat{\delta}_m\|}{2c_0} \hat{\gamma}_m \right). \tag{11.38}$$

These are nearly twice the size of the uncorrected values.

Table 11.4 Ebay output for the Michigan lung cancer study. Correlation error estimates $\hat{\alpha}_{\text{cor}}$ are much more pessimistic, as confirmed by cross-validation.

Step	Index	z-value	$\hat{\delta}$	$\hat{\alpha}$	$\hat{\alpha}_{\text{cor}}$
1	3144	4.62	3.683	.3290	.329
2	2446	4.17	3.104	.2813	.307
3	4873	4.17	3.104	.2455	.256
4	1234	3.90	2.686	.2234	.225
5	621	3.77	2.458	.2072	.213
6	676	3.70	2.323	.1942	.228
7	2155	3.69	2.313	.1824	.230
8	3103	3.60	2.140	.1731	.236
9	1715	3.58	2.103	.1647	.240
10	452	3.54	2.028	.1574	.243
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
193	3055	2.47	.499	.0519	.359
194	1655	-2.21	-.497	.0518	.359
195	2455	2.47	.496	.0517	.359
196	3916	2.47	.496	.0516	.359
197	4764	2.47	.495	.0515	.359
198	1022	-2.20	-.492	.0514	.359
199	1787	-2.19	-.490	.0513	.360
200	901	-2.18	-.486	.0512	.360

Table 11.4 shows Ebay output for the Michigan lung cancer study, a microarray experiment with $N = 5217$ genes and $n = 86$ subjects, $n_1 = 62$ “good outcomes” and $n_2 = 24$ “poor outcomes”. Correlation problems are much more severe here. Ebay stopped after $m = 200$ steps (the default stopping rule) without $\hat{\alpha}$ reaching the target value $\alpha_0 = 0.025$. The correlation-corrected errors $\hat{\alpha}_{\text{cor}}$ are much more pessimistic, increasing after the first six steps, eventually to $\hat{\alpha}_{\text{cor}} = 0.36$. A cross-validation error rate of $\hat{\alpha}_{\text{CV}} = 0.37$ confirmed the pessimism.

We could employ more elaborate recipes for selecting an empirical Bayes prediction rule, for example “Use the rule corresponding to the minimum value of $\hat{\alpha}_{\text{cor}}$ in the first 200 steps.” (This would select the rule based on the first five genes in Table 11.4.) If Ebay were programmed to follow such recipes automatically, which it isn’t, we could then get honest cross-validation error rates $\hat{\alpha}_{\text{CV}}$ just as before.

11.3 Prediction and Local False Discovery Rates

The local false discovery rate $\text{fdr}(z)$ (5.2) is

$$\text{fdr}(z) = \pi_0 \varphi(z) / f(z) \quad (11.39)$$

under the theoretical null distribution $f_0(z) = \varphi(z)$ (2.6). Comparing (11.39) with (11.25) and (11.27) gives the following.

Corollary 11.3 *Under the normal hierarchical model (11.22), δ has posterior mean and variance*

$$E\{\delta|z\} = -\frac{d}{dz} \log \text{fdr}(z) \quad \text{and} \quad \text{var}\{\delta|z\} = -\frac{d^2}{dz^2} \log \text{fdr}(z) \quad (11.40)$$

when $\text{fdr}(z)$ is calculated using the theoretical null (11.39).

There is something a little surprising about this result. It seemingly makes sense that genes with low false discovery rates should be the ones utilized in prediction rules, but (11.40) shows that this is not exactly true: large values of $\tilde{\delta}_i = E\{\delta_i|z_i\}$, the Bayes effect size estimate, depend on the rate of change of $\log \text{fdr}(z_i)$, not on $\text{fdr}(z_i)$ itself. Small values of $\text{fdr}(z_i)$ usually correspond to large values of $|\tilde{\delta}_i|$, but this doesn’t have to be the case.

Let $A(x)$ be the area under the curve $E\{\delta|z\}$ between 0 and x ,

$$A(x) = \int_0^x E\{\delta|z\} dz. \quad (11.41)$$

The first equation in (11.40) then gives the following.

Corollary 11.4 Under (11.22), (11.39),

$$\frac{\text{fdr}(x)}{\text{fdr}(0)} = e^{-A(x)}. \quad (11.42)$$

(This remains true for $x < 0$ with the definition $A(x) = -\int_x^0 E\{\delta|z\}dz$.)

Looking at $\hat{E}\{\delta|z\}$ for the prostate data, Figure 11.1, we can see that $A(x)$ is nearly zero for $x < 2$ but then increases roughly quadratically with x , driving down $\text{fdr}(x)$ according to (11.42).

Suppose that the appropriate null hypothesis distribution is $\mathcal{N}(\delta_0, \sigma_0^2)$ as in (6.9), rather than $\mathcal{N}(0, 1)$. We can change the normal hierarchical model (11.22) to

$$\delta \sim g(\cdot) \quad \text{and} \quad \frac{z - \delta_0}{\sigma_0} \Big| \delta \sim \mathcal{N}(\delta, 1) \quad (11.43)$$

so that δ still represents effect size on a $\mathcal{N}(0, 1)$ scale. The case $\delta = 0$ corresponds to null density $f_0(z) = \varphi((z - \delta_0)/\sigma_0)/\sigma_0$, giving $\text{fdr}(z) = \pi_0 f_0(z)/f(z)$, where the marginal density $f(z)$ is

$$f(z) = \int_{-\infty}^{\infty} \frac{1}{\sigma_0} \varphi\left(\frac{z - \delta_0 - \delta}{\sigma_0}\right) g(\delta) d\delta. \quad (11.44)$$

Differentiating $\log \text{fdr}(z)$ gives

$$-\frac{d}{dz} \log \text{fdr}(z) = \frac{z - \delta_0}{\sigma_0^2} + l'(z) \quad \left[l'(z) = \frac{d}{dz} \log f(z) \right] \quad (11.45)$$

while an exponential family calculation similar to that for Theorem 11.1 yields a transformed version of (11.28),

$$E\{\delta|z\} = \frac{z - \delta_0}{\sigma_0} + \sigma_0 l'(z) \quad \text{and} \quad \text{var}\{\delta|z\} = 1 + \sigma_0^2 l''(z). \quad (11.46)$$

So $d \log \text{fdr}(z)/dz = E\{\delta|z\}/\sigma_0$, and the transformed version of Corollary 11.4 is

$$\frac{\text{fdr}(x)}{\text{fdr}(\delta_0)} = e^{-A_0(x)} \quad \left[A_0(x) = \int_{\delta_0}^x E\{\delta|z\} dz / \sigma_0 \right]. \quad (11.47)$$

Exercise 11.6 Verify (11.45) and (11.46). Why does (11.46) differ from (11.35)?

Figure 11.2 compares $E\{\delta|z\}$ from models (11.22) and (11.43) for the prostate and leukemia data. Here we have chosen (δ_0, σ_0) in (11.43) to be the MLE empirical null estimates of Section 6.3,

$$\text{prostate } (\delta_0, \sigma_0) = (0.003, 1.06) \quad \text{leukemia} = (0.094, 1.68). \quad (11.48)$$

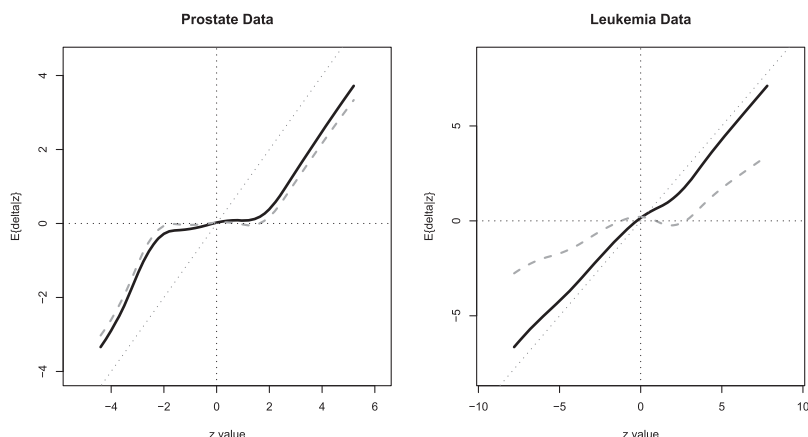


Figure 11.2 $E\{\delta|z\}$ from the normal hierarchical model (11.22), solid curves; and from the transformed version (11.43), dashed curves, with (δ_0, σ_0) the MLE empirical null values (11.48). Changing from the theoretical to empirical null flattens $E\{\delta|z\}$ near $z = 0$.

Note that the slight slope of $\hat{E}\{\delta|z\}$ between -2 and 2 seen in Figure 11.1 for the prostate data is flattened by changing from $(\delta_0, \sigma_0) = (0, 1)$ to the empirical null values. The flattening effect is much more dramatic for the leukemia data. Area $A_0(x)$ is much smaller than $A(x)$ and (11.47) is bigger than (11.42), demonstrating in geometric terms why it is more difficult in this case to get small values of fdr with the empirical null.

Large-scale hypothesis testing usually begins with the assumption that most of the effects are zero, or at least small, as in (2.7)–(2.8), which suggests that $E\{\delta|z\}$ should be near zero and flat for central values of z . The main point of Figure 11.2 is diagnostic: $\hat{E}\{\delta|z\} = z + \hat{l}'(z)$ passing through $z = 0$ at a sharp angle is a warning against casual use of the theoretical null.

Figure 11.3 concerns another, more vivid, geometric representation of the relationship between false discovery rate and effect size. Let $e(z)$ equal $E\{\delta|z\}$ in the original hierarchical model (11.22),

$$e(z) = z + l'(z) \quad (11.49)$$

(11.29), and define (δ_0, σ_0) from $l(z) = \log(f(z))$ according to

$$\delta_0 : l'(\delta_0) = 0 \quad \text{and} \quad \sigma_0 = 1/(-l''(\delta_0))^{1/2}; \quad (11.50)$$

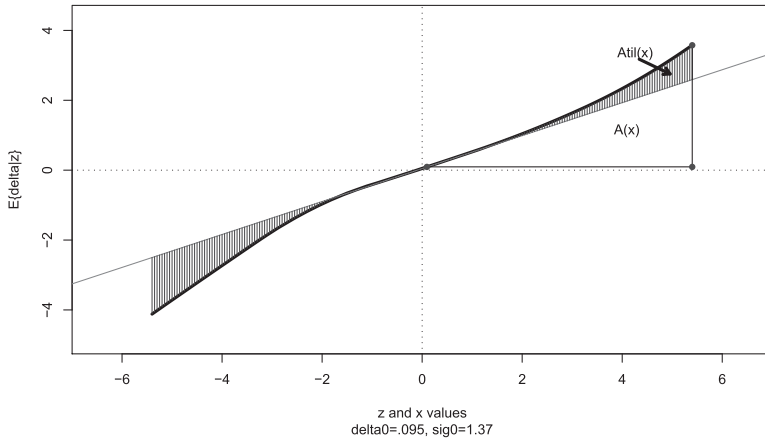


Figure 11.3 Corollary 11.5 illustrated for the police data of Figure 6.1c; hatched area $\tilde{A}(x)$ determines the local false discovery rate based on empirical null (11.53). It is much smaller than $A(x)$, the area of the outlined region (including $\tilde{A}(x)$) that determines the theoretical null fdr (11.42). (Heavy curve is $E\{\delta|z\}$ calculated according to the original hierarchical model (11.22).)

(δ_0, σ_0) are the central matching estimates (6.11). The tangent line to $e(z)$ at $z = \delta_0$ is

$$e_{\tan}(z) = e(\delta_0) + e(\delta_0)'(z - \delta_0) \quad [e(z)' = de(z)/dz]. \quad (11.51)$$

Corollary 11.5 Let $\tilde{A}(x)$ be the area between $e(z)$ and $e_{\tan}(z)$,

$$\tilde{A}(x) = \int_{\delta_0}^x [e(z) - e_{\tan}(z)] dz. \quad (11.52)$$

Then

$$\frac{\widetilde{\text{fdr}}(x)}{\widetilde{\text{fdr}}(\delta_0)} = e^{-\tilde{A}(x)} \quad (11.53)$$

where $\widetilde{\text{fdr}}(z)$ is the local false discovery rate $\pi_0 f_0(z)/f(z)$ with $f_0(z)$ the central matching empirical null distribution.⁴

(Proof at the end of this section.)

⁴ The locfdr central matching algorithm actually applies calculations (11.50) to a local quadratic fit to $\log \hat{f}(z)$. It gave $(\hat{\delta}_0, \hat{\sigma}_0) = (0.04, 1.45)$ for the police data, rather than $(0.09, 1.37)$ from (11.50).

Exercise 11.7 Show that for (δ_0, σ_0) , the central matching values (11.50), the transformed version of $E\{\delta|z\}$ in (11.46) satisfies

$$\left. \frac{dE\{\delta|z\}}{dz} \right|_{z=\delta_0} = 0 \quad (11.54)$$

(so that $E\{\delta|z\}$ is perfectly flattened near $z = 0$).

One thing is noticeably absent from Corollary 11.4 and Corollary 11.5: π_0 , the prior null probability in $\text{fdr}(z) = \pi_0 f_0(z)/f(z)$. Its place is taken by $\text{fdr}(0)$ or $\text{fdr}(\delta_0)$ in (11.42), (11.53). For the two-groups model (5.2),

$$\text{fdr}(0) = 1/(1 + \text{OR}(0)) \quad [\text{OR}(z) = (\pi_1 f_1(z))/(\pi_0 f_0(z))], \quad (11.55)$$

$\text{OR}(z)$ being the posterior odds ratio for *non-null* given z . Under the *zero assumption* (4.44) we employed to estimate π_0 , $\text{OR}(0) = 0$ and $\text{fdr}(0) = 1$. It is quite reasonable to interpret the corollaries as saying

$$\text{fdr}(x) = e^{-A(x)} \quad \text{and} \quad \widetilde{\text{fdr}}(x) = e^{-\tilde{A}(x)}, \quad (11.56)$$

thereby finessing the problem of estimating π_0 .

$A(x)$ is easier to estimate than $\tilde{A}(x)$ in Figure 11.3. The latter requires estimating the lower tangent line, adding variability to the assessment of a smaller quantity. This results in less accurate fdr estimates when based on empirical null distributions, as seen in Table 7.3. Of course there is really no choice in the police data example, where the theoretical null is unrealistic.

Proof of Corollary 11.5 From $e(z) = z + l'(z)$ (11.49) and $e(z)' = 1 + l''(z)$, (11.50) gives

$$e_{\tan}(z) = \delta_0 + \left(1 - 1/\sigma_0^2\right)(z - \delta_0) \quad (11.57)$$

and

$$\tilde{e}(z) \equiv e(z) - e_{\tan}(z) = \frac{z - \delta_0}{\sigma_0^2} + l'(z). \quad (11.58)$$

But this equals $-d \log \text{fdr}(z)/dz$ (11.45), yielding (11.53). $\square$

11.4 Effect Size Estimation

The traditional purpose of simultaneous significance testing has been to identify non-null cases, usually assumed to be a small subset of all N possibilities in large-scale situations. Here we take up a more ambitious goal: to assess effect sizes for those so identified, that is, to estimate how far away non-null cases lie from the null hypothesis. Not all non-null cases

are created equal: those with large effect sizes δ_i or $-\delta_i$ are the most useful for prediction, and most likely to play key roles in an ongoing process of scientific discovery.

We now combine the normal hierarchical model (11.22) with the two-groups model (5.1) by assuming that the prior density $g(\delta)$ has an atom of probability π_0 at $\delta = 0$, leaving probability $\pi_1 = 1 - \pi_0$ for the non-zero values,

$$g(\delta) = \pi_0 \Delta_0(\delta) + \pi_1 g_1(\delta) \quad \text{and} \quad z|\delta \sim \mathcal{N}(\delta, 1). \quad (11.59)$$

Here $\Delta_0(\cdot)$ represents a delta function at 0 while $g_1(\cdot)$ is the prior density for non-null δ values; (11.59) gives a two-groups model (5.1) with $f_0(z) = \varphi(z)$, the $\mathcal{N}(0, 1)$ density, and

$$f_1(z) = \int_{-\infty}^{\infty} \varphi(z - \delta) g_1(\delta) d\delta. \quad (11.60)$$

We also define J to be the indicator of the null status of a case,

$$J = \begin{cases} 0 & \text{null } (\delta = 0) \\ 1 & \text{non-null } (\delta \neq 0) \end{cases} \quad (11.61)$$

(the reverse of the indicator I used previously).

By definition,

$$\text{fdr}(z) = \Pr\{J = 0|z\}, \quad (11.62)$$

the conditional probability of nullity given z . For any value j ,

$$\begin{aligned} E\{\delta^j|z\} &= E\{\delta^j|z, J = 0\} \text{fdr}(z) + E\{\delta^j|z, J = 1\} (1 - \text{fdr}(z)) \\ &= E\{\delta^j|z, J = 1\} (1 - \text{fdr}(z)) \end{aligned} \quad (11.63)$$

since $\delta = 0$ if $J = 0$.

For convenient notation, let

$$E_1(z) = E\{\delta|z, J = 1\} \quad \text{and} \quad \text{var}_1(z) = \text{var}\{\delta|z, J = 1\}, \quad (11.64)$$

the conditional mean and variance for a non-null case.

Exercise 11.8 Use (11.63) to show that

$$E_1(z) = \frac{E\{\delta|z\}}{1 - \text{fdr}(z)} \quad \text{and} \quad \text{var}_1(z) = \frac{\text{var}\{\delta|z\}}{1 - \text{fdr}(z)} - \text{fdr}(z)E_1(z)^2. \quad (11.65)$$

Combined with (11.28), this yields

$$E_1(z) = \frac{z + l'(z)}{1 - \text{fdr}(z)} \quad \text{and} \quad \text{var}_1(z) = \frac{1 + l''(z)}{1 - \text{fdr}(z)} - \text{fdr}(z)E_1(z)^2. \quad (11.66)$$

Plugging in estimates of $l(z)$ and $\widehat{\text{fdr}}(z)$ then gives $\hat{E}_1(z)$ and $\widehat{\text{var}}_1(z)$. Figure 11.4 shows the resulting 90% empirical Bayes posterior bands for the right tail of the prostate data,

$$\hat{E}_1(z) \pm 1.645 \widehat{\text{var}}_1(z)^{1/2}. \quad (11.67)$$

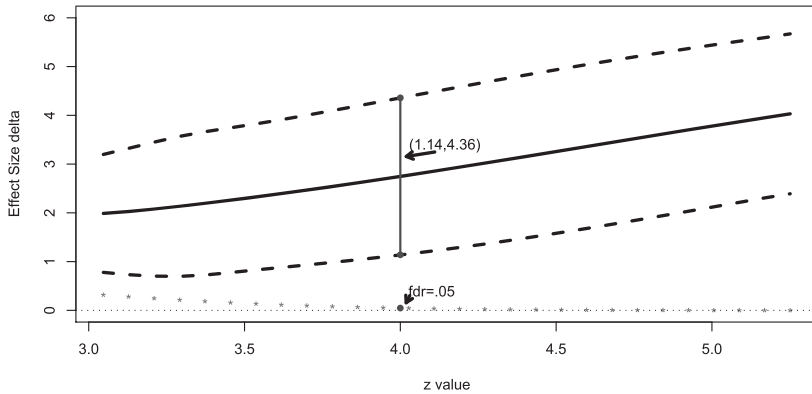


Figure 11.4 Estimated 90% empirical Bayes posterior band for a non-null effect (11.67), prostate data, and estimated local false discovery rate $\widehat{\text{fdr}}(z)$. At $z = 4$, $\widehat{\text{fdr}} = 0.05$, and the 90% interval is (1.14, 4.36).

At $z = 4$ we have $\widehat{\text{fdr}}(z) = 0.05$, $\hat{E}_1(z) = 2.75$, and $\widehat{\text{var}}_1(z)^{1/2} = 0.98$, so interval (11.67) is (1.14, 4.36). The literal interpretation is that, given $z = 4$, there is a 5% chance that $\delta = 0$ and, if not, a 90% chance that δ lies in (1.14, 4.36). In other words, δ is positive with high probability and, if so, the effect size is probably substantial.

From a traditional point of view, it seems strange that our assessment of δ is split into two parts: a point probability at zero and an interval that does not contain zero. But this is a natural consequence of the “mixed” prior $g(\delta)$ in (11.59) which combines an atom at zero with a possibly diffuse, non-zero component. This reinforces the point made in previous chapters that large-scale significance testing is more than just a collection of individual tests done at the same time. Estimation and prediction are also involved, as indirect evidence of an empirical Bayes nature, “learning from the experience of others,” makes itself felt.

A version of Theorem 11.1 and Corollary 11.3 is available that applies directly to the non-null posterior distribution $g_1(\delta) = g(\delta|z, J = 1)$.

Theorem 11.6 Under model (11.59), $g_1(\delta|z)$ is an exponential family,

$$g_1(\delta) = e^{\delta z - \psi_1(z)} \left[e^{-\delta^2/2} g_1(\delta) \right] \quad (11.68)$$

where

$$\psi_1(z) = \log \left\{ \frac{1 - \text{fdr}(z)}{\text{fdr}(z)} \frac{\pi_1}{\pi_0} \right\}. \quad (11.69)$$

Proof Bayes rule says that $g_1(\delta|z) = \varphi(z - \delta)g_1(\delta)/f_1(z)$, yielding

$$g_1(\delta|z) = e^{\delta z - \log\{f_1(z)/\varphi(z)\}} \left[e^{-\delta^2/2} g_1(\delta) \right]. \quad (11.70)$$

□

Exercise 11.9 Complete the proof by using $\text{fdr}(z) = \pi_0\varphi(z)/f(z)$ to show that

$$\frac{f_1(z)}{\varphi(z)} = \frac{\pi_0}{\pi_1} \frac{1 - \text{fdr}(z)}{\text{fdr}(z)}. \quad (11.71)$$

Differentiating $\psi_1(z)$ yields non-null posterior cumulants of δ given z , the j th cumulant being

$$- \frac{d^j}{dz^j} \log \left\{ \frac{\text{fdr}(z)}{1 - \text{fdr}(z)} \right\}; \quad (11.72)$$

$j = 1$ and 2 are the non-null conditional mean and variance (as compared with (11.40)), agreeing of course with expressions (11.65).

False coverage rate control

A more classically oriented frequentist approach to effect size estimation aims to control FCR, the false coverage rate. We are given a method of discovering non-null cases, for example the BH(q) algorithm of Chapter 4, and also of assigning confidence intervals for the effect sizes δ_i of the discovered cases. The method's FCR is the *expected* proportion of discovered cases in which δ_i is not in its assigned confidence interval.

FCR control is a laudable criterion, aimed against the bad practice of assigning the usual confidence intervals to those cases selected as non-null — say 5.29 ± 1.645 for δ_{610} in Table 11.2 — which completely ignores selection bias.

The BY(q) FCR control algorithm⁵ works as follows: let $\text{CI}_z(p)$ be the usual two-sided normal confidence interval of coverage probability $1 - p$,

$$\text{CI}_z(p) = z \pm \Phi^{-1} \left(1 - \frac{p}{2} \right) \quad (11.73)$$

⁵ Benjamini–Yekutieli; see Notes at this chapter's end.

(so $p = 0.10$ corresponds to $z \pm 1.645$). Perform the $BH(q)$ FDR control algorithm of Section 4.2 on $p_1, p_2, \dots, p_N$ with $p_i = 2\Phi(-|z_i|)$, declaring, say, R of the z_i non-null. To each corresponding δ_i assign the confidence interval

$$CI_{z_i}\left(\frac{Rq}{N}\right) = z_i \pm \Phi^{-1}\left(1 - \frac{Rq}{2N}\right). \quad (11.74)$$

Then, assuming the z_i are *independent* normal unbiased estimates of their effect sizes δ_i ,

$$z_i \stackrel{\text{ind}}{\sim} \mathcal{N}(\delta_i, 1) \quad i = 1, 2, \dots, N, \quad (11.75)$$

the false coverage rate will be no greater than q .

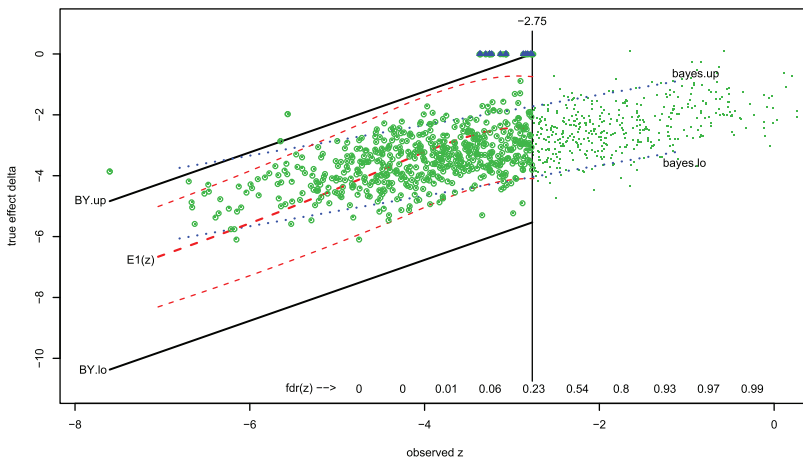


Figure 11.5 False coverage rate example: $N = 10\,000$ pairs (z_i, δ_i) from model (11.75)–(11.76). Green points, the 1000 non-null cases, $\delta_i \sim \mathcal{N}(-3, 1)$; circled points, those 563 declared non-null by $BH(0.10)$ algorithm. Heavy black lines, $BY(0.10)$ FDR control confidence limits (11.74); blue dotted lines, Bayes 90% limit for non-null cases (11.80); red dashed lines, $\hat{E}_1(z)$ (11.65) and nominal 90% empirical Bayes limits (11.67). Bottom numbers list $fdr(z)$ for model (11.75)–(11.76). Dark triangles are 14 null cases ($\delta_i = 0$) having $z_i < -2.75$; 26 more, not pictured, had $z_i > 2.75$.

Figure 11.5 shows an application of the $BY(q)$ algorithm, $q = 0.10$, to a simulated data set in which $N = 10\,000$ independent (δ_i, z_i) pairs were

generated as in (11.59), with

$$g(\delta) = 0.90\Delta_0(\delta) + 0.10\varphi_{-3,1}(\delta). \quad (11.76)$$

That is, 90% of the effects were zero and 10% $\mathcal{N}(-3, 1)$. The BH(0.10) rule declared $R = 603$ cases non-null, those having $|z_i| \geq 2.75$. Using (11.74), the BY(q) algorithm assigned confidence intervals

$$z_i \pm 2.75 \quad (11.77)$$

to the 603 cases declared non-null.

Exercise 11.10 Why is it *not* a coincidence that the BH cutoff $|z_i| \geq 2.75$ is nearly the same as the half-width 2.75 in (11.77)?

Because this is a simulation, we can see how well the BY algorithm performed. Figure 11.5 plots (z_i, μ_i) points for the 1000 *actual* non-null cases, those having $\delta_i \sim \mathcal{N}(-3, 1)$; 563 of these were *declared* non-null, those with $z_i \leq -2.75$. There were 40 false discoveries, those having $\delta_i = 0$, 14 with $z_i \leq -2.75$ and 26 with $z_i \geq 2.75$ (not pictured).

The first thing to notice is that the FCR property is satisfied: only 43 of the $R = 603$ confidence intervals fail to cover δ_i , those for the 40 false discoveries and three of 563 true discoveries, giving $43/603 = 0.07$, substantially less than $q = 0.10$. However, the second thing is that the intervals (11.77) are frighteningly wide — about 67% longer than the individual 90% intervals $z_i \pm 1.645$ — and poorly centered, particularly at the left where all the δ_i fall in the upper halves of their intervals, as seen from the heavy lines for the BY confidence limits.

An interesting comparison is with Bayes rule applied to (11.75)–(11.76). This yields

$$\Pr\{\delta = 0|z_i\} \equiv \text{fdr}(z_i) = \frac{0.90\varphi(z)}{0.90\varphi(z) + 0.10\varphi_{-3,\sqrt{2}}(z)} \quad (11.78)$$

$(\varphi_{\delta,\sigma}(z) = \varphi((z - \delta)/\sigma)/\sigma)$ and non-null posterior density

$$g_1(\delta_i|z_i) \equiv g(\delta_i|z_i, J_i = 1) \sim \mathcal{N}\left(\frac{z_i - 3}{2}, \frac{1}{2}\right). \quad (11.79)$$

That is, δ_i is null with probability $\text{fdr}(z_i)$ and $\mathcal{N}((z_i - 3)/2, 1/2)$ with probability $1 - \text{fdr}(z_i)$. The blue dotted lines indicate the 90% posterior intervals given that δ_i is non-null,

$$(z_i - 3)/2 \pm 1.645/\sqrt{2}, \quad (11.80)$$

now a factor $\sqrt{2}$ shorter than $z_i \pm 1.645$.

The Bayes intervals and the frequentist $BY(q)$ intervals are pursuing the same goal, to include the effect sizes δ_i with 90% certainty. At $z = -2.75$, the $BY(q)$ assessment is $\Pr\{\delta \in (-5.50, 0)\} = 0.90$, while Bayes rule states that $\delta = 0$ with probability $\text{fdr}(-2.75) = 0.23$, and otherwise $\delta \in (-4.04, -1.71)$ with probability 0.90. This kind of bifurcated assessment, as in Figure 11.4, is inherent to model (11.59). A principal cause of $BY(q)$'s oversized intervals comes from using a connected set to describe a disconnected situation.

As described in Chapter 4, the $BH(q)$ FDR control algorithm enjoyed both frequentist and Bayesian support, and both philosophies played roles in the subsequent chapters' empirical Bayes development. The same cannot be said for FCR control and the $BY(q)$ algorithm. Frequentist methods like $BY(q)$ enjoy the considerable charm of exact error control, without requiring prior distributions, but the discrepancy evident in Figure 11.5 is disconcerting.

The red dashed lines in Figure 11.5 are the 90% empirical Bayes intervals (11.67). (Actually using $\widehat{\text{var}}\{\delta|z\} = 1 + \hat{l}''(z)$ (11.28), rather than $\widehat{\text{var}}_1(z)$. These are nearly the same for $\widehat{\text{fdr}}(z)$ small, while the former is a more stable estimator.) These are closer to the Bayes intervals, and $\widehat{\text{fdr}}(z)$ is close to the true $\text{fdr}(z)$, e.g., $\text{fdr}(-2.75) = 0.27$, but the results are far from perfect.

There is a hierarchy of estimation difficulties connected with the methods of this chapter: $\text{fdr}(z)$ is the easiest to estimate since it only requires assessing the marginal density $f(z)$ or its logarithm $l(z)$; $E\{\delta|z\} = z + l'(z)$ requires estimating the *slope* of $l(z)$, while $\text{var}\{\delta|z\} = 1 + l''(z)$ requires its curvature. The Poisson regression estimates of Section 5.2 are quite dependable for $\text{fdr}(z)$, adequate for $E\{\delta|z\}$, and borderline undependable for $\text{var}\{\delta|z\}$. Empirical Bayes assessment of effect size is a worthy goal, and we can hope for the development of a more stable methodology.

11.5 The Missing Species Problem

We conclude with perhaps the first empirical Bayes success story, the solution of the "missing species problem." The original motivating data set is shown in Table 11.5. Alexander Corbet, a naturalist, spent two years in the early 1940s trapping butterflies in Malaysia (then Malaya). Let

$$n_j = \#\{\text{species trapped exactly } j \text{ times in two years}\}. \quad (11.81)$$

The table shows $n_1 = 118$, this being the number of species so rare that Corbet captured just one each; $n_2 = 74$ species were trapped twice each, n_3 three times each, etc.

Table 11.5 *Corbet’s Malaysian butterfly data; n_j is the number of butterfly species captured exactly j times in two years of trapping: 118 species were trapped just once each, 74 two times each, etc.*

j	1	2	3	4	5	6	7	8	9	10	11	12	...
n_j	118	74	44	24	29	22	20	19	20	15	12	14	...

Corbet then asked a very interesting prediction question: if he spent an additional one year trapping — half as long as he had already been there — how many *new* species could he expect to capture? One reason the question is interesting is that it seems impossible to answer. It refers to n_0 , the number of Malaysian butterfly species *not* previously captured. And of course n_0 is missing from Table 11.5.

Fortunately, Corbet asked the right person: R. A. Fisher, the founding father of modern statistical theory. We will return later to Fisher’s solution, which involves a parametric empirical Bayes argument, but first we present a non-parametric solution developed in the 1950s. Let $\mathcal{S}$ be the set of all Malaysian butterfly species and let $S = \#\mathcal{S}$ be their number ($S \geq 435$ for Corbet’s problem, the total of Table 11.5). The key assumption is that each species s is trapped according to a Poisson process with intensity parameter λ_s ,

$\lambda_s =$ Expected number of species s captured per unit time. (11.82)

The time unit is two years in Corbet’s example.

Letting x_s be the number of times species s is trapped in the original one unit of observation time, x_s has Poisson probability density function

$p_s(j) = \Pr\{x_s = j\} = e^{-\lambda_s} \lambda_s^j / j!$ (11.83)

for $j = 0, 1, 2, \dots$. The expectation η_j of n_j is then given by

$\eta_j = E\{n_j\} = \sum_s e^{-\lambda_s} \lambda_s^j / j! = S \int_0^\infty e^{-\lambda} \lambda^j / j! dG(\lambda)$ (11.84)

where $G(\lambda)$ is the (unobservable) empirical cdf of the λ_s values,

$G(\lambda) = \#\{\lambda_s \leq \lambda\} / S.$ (11.85)

The essence of Poisson model (11.82) is that species s is trapped independently in non-overlapping time intervals. In particular, for a new trapping period of length t , the number of times species s will be captured

follows a $\text{Poi}(\lambda_s t)$ distribution, independent of its original capture number $x_s \sim \text{Poi}(\lambda_s)$. Therefore species s has probability

$$q_s(t) = e^{-\lambda_s} (1 - e^{-\lambda_s t}) \quad (11.86)$$

of *not being seen* during the original period and *being seen* in the new trapping period.

Adding up over species, $\mu(t)$, the expected number of new species seen in succeeding time t is

$$\begin{aligned} \mu(t) &= \sum_s q_s(t) = S \int_0^\infty e^{-\lambda} (1 - e^{-\lambda t}) dG(\lambda) \\ &= S \int_0^\infty e^{-\lambda} [\lambda t - (\lambda t)^2/2! + (\lambda t)^3/3! + \dots] dG(\lambda), \end{aligned} \quad (11.87)$$

the last line following by Taylor expansion of $1 - \exp(-\lambda t)$. Comparing (11.87) with (11.84) yields an intriguing formula,

$$\mu(t) = \eta_1 t - \eta_2 t^2 + \eta_3 t^3 \dots \quad (11.88)$$

Each η_j is unbiasedly estimated by its corresponding count n_j (11.81), giving an unbiased estimate of $\mu(t)$,

$$\hat{\mu}(t) = n_1 t - n_2 t^2 + n_3 t^3 \dots, \quad (11.89)$$

not involving the zero count n_0 .

Now we can answer Corbet's question: for a proposed additional trapping period of one year, i.e., $t = 1/2$ as long as the original period, our prediction for the number of new species captured is⁶

$$\hat{\mu}(t) = 118 \left(\frac{1}{2} \right) - 74 \left(\frac{1}{4} \right) + 44 \left(\frac{1}{8} \right) - 24 \left(\frac{1}{16} \right) \dots = 45.2. \quad (11.90)$$

How accurate is $\hat{\mu}(t)$? Just as for the empirical Bayes estimates of Chapter 2, the unbiasedness of $\hat{\mu}(t)$ does not depend on independence among the species-capture values x_s , but its accuracy does. A useful bound on the variance of $\hat{\mu}(t)$, developed next, is available assuming independence,

$$x_s \stackrel{\text{ind}}{\sim} \text{Poi}(\lambda_s) \quad \text{for } s \in \mathcal{S}. \quad (11.91)$$

Let $\mathbf{p}_s = (p_s(0), p_s(1), p_s(2), \dots)'$ be the vector⁷ of Poisson probabilities (11.83) and $\mathbf{I}_s$ the indicator vector

$$\mathbf{I}_s = (0, 0, \dots, 0, 1, 0, \dots)' \quad (11.92)$$

⁶ Perhaps a discouraging figure; it is not clear if Corbet continued his stay.

⁷ Theoretically, $\mathbf{p}_s$ and $\mathbf{I}_s$ have an infinite number of components, but that does not affect the calculations that follow.

with 1 in the j th place if $x_s = j$. A standard multinomial calculation shows that $\mathbf{I}_s$ has expectation $\mathbf{p}_s$ and covariance matrix

$$\text{cov}(\mathbf{I}_s) = \text{diag}(\mathbf{p}_s) - \mathbf{p}_s \mathbf{p}_s', \quad (11.93)$$

“diag” denoting a diagonal matrix with the indicated entries (so $\text{var}(p_s(j)) = p_s(j)(1 - p_s(j))$, the usual binomial formula).

The vectors $\mathbf{n} = (n_0, n_1, n_2, \dots)'$ and $\boldsymbol{\eta} = (\eta_0, \eta_1, \eta_2, \dots)'$ are given by

$$\mathbf{n} = \sum_s \mathbf{I}_s \quad \text{and} \quad \boldsymbol{\eta} = \sum_s \mathbf{p}_s. \quad (11.94)$$

Assuming independence,

$$\text{cov}(\mathbf{n}) = \sum_s \text{cov}(\mathbf{I}_s) = \text{diag}(\boldsymbol{\eta}) - \sum_s \mathbf{p}_s \mathbf{p}_s'. \quad (11.95)$$

Defining

$$\mathbf{u}(t) = (0, t, -t^2, t^3, \dots)', \quad (11.96)$$

(11.89) gives

$$\hat{\mu}(t) = \sum_j u(t)_j n_j = \mathbf{u}(t)' \mathbf{n} \quad (11.97)$$

and

$$\begin{aligned} \text{var}(\hat{\mu}(t)) &= \mathbf{u}(t)' \text{cov}(\mathbf{n}) \mathbf{u}(t) \\ &= \mathbf{u}(t)' \text{diag}(\boldsymbol{\eta}) \mathbf{u}(t) - \sum_s (\mathbf{u}(t)' \mathbf{p}_s)^2. \end{aligned} \quad (11.98)$$

We see that an upper bound on $\text{var}(\hat{\mu}(t))$ is

$$\text{var}(\hat{\mu}(t)) \leq \mathbf{u}(t)' \text{diag}(\boldsymbol{\eta}) \mathbf{u}(t) = \sum_{j \geq 1} \eta_j t^{2j}. \quad (11.99)$$

Substituting n_j for η_j provides an approximation for the standard deviation of $\hat{\mu}(t)$,

$$\widehat{\text{sd}}\{\hat{\mu}(t)\} = \sum_{j \geq 1} n_j t^{2j}. \quad (11.100)$$

This yields $\widehat{\text{sd}} = 9.3$ for Corbet's data, so the coefficient of variation of estimate 45.2 is about 20%. The term we are ignoring in (11.98), $\sum_s (\mathbf{u}(t)' \mathbf{p}_s)^2$, is usually small. Without it, (11.99) gives $\text{var}(\hat{\mu}(t))$ under the Poisson model $n_j \overset{\text{ind}}{\sim} \text{Poi}(\eta_j)$, not unreasonable in its own right.

Exercise 11.11 Show that the ignored term equals

$$\sum_s (\mathbf{u}(t)' \mathbf{p}_s)^2 = S \int_0^\infty q_\lambda(t)^2 dG(\lambda) \quad (11.101)$$

$$[q_\lambda(t) = e^{-\lambda} (1 - e^{-\lambda t})].$$

For an additional trapping period of two years, i.e., $t = 1$, (11.89), (11.100) predict 75 ± 20 new species. The series in (11.89) diverges for $t > 1$, invalidating the predictor $\hat{\mu}(t)$. Fisher's original approach to the missing species problem avoids this limitation at the expense of assuming that the intensity cdf (11.85) is obtained from a gamma density function for $\lambda \geq 0$,

$$g(\lambda) = c(\alpha, \beta) \lambda^{\alpha-1} e^{-\lambda/\beta} \quad [c(\alpha, \beta) = (\beta^\alpha \Gamma(\alpha))^{-1}]. \quad (11.102)$$

The previous expressions now take on specific forms; for instance,

$$\eta_j = S \frac{\Gamma(\alpha + j)}{j! \Gamma(\alpha)} \quad (11.103)$$

for (11.84). Some algebra yields Fisher's prediction formula:

Exercise 11.12 Show that the gamma assumption (11.102) results in

$$\mu(t) = \eta_1 \{1 - 1/(1 + \gamma t)^\alpha\} / (\gamma \alpha) \quad [\gamma = \beta/(1 + \beta)]. \quad (11.104)$$

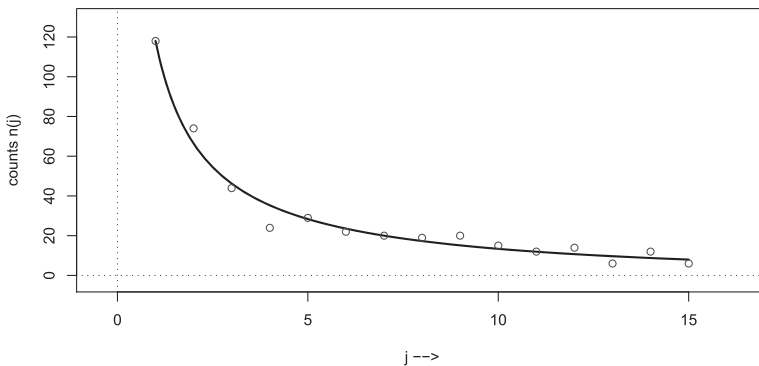


Figure 11.6 Solid curve is Fisher's negative binomial estimates $\hat{\eta}_j$ (11.105), compared to counts n_j from Corbet's Table 11.5; $\hat{\eta}_1 = 118$, $\hat{\alpha} = 0.166$, $\hat{\beta} = 27.06$.

Formula (11.103) can be re-expressed in *decapitated negative binomial*

form

$$\eta_j = \eta_1 \frac{\Gamma(\alpha + j)}{j! \Gamma(\alpha + 1)} \gamma^{j-1}, \tag{11.105}$$

decapitation referring to the elimination of η_0 . Maximum likelihood estimation in the model $n_j \overset{\text{ind}}{\sim} \text{Poi}(\eta_j)$ gave MLE values $\hat{\alpha} = 0.166$ and $\hat{\beta} = 27.06$. Figure 11.6 compares Fisher’s estimate $\hat{\eta}_j$ from (11.105) (taking $\hat{\eta}_1 = n_1$) with n_j from Table 11.5. The fit is not wonderful, but the results are similar to those obtained non-parametrically: $\hat{\mu}(0.5) = 46.6$, $\hat{\mu}(1) = 78.1$.

We can go as far into the future as we wish now — for example, $\hat{\mu}(2) = 120.4$ expected new species for four additional trapping years — but the predictions become increasingly speculative. The crucial function

$$q_\lambda(t) = e^{-\lambda} \left(1 - e^{-\lambda t}\right) \tag{11.106}$$

which determines the probability that a species of intensity λ will be a new capture (11.86), concentrates ever nearer zero as t increases. It is easy to show that $q_\lambda(t)$ is maximized at

$$\lambda_{\max} = (\log(t + 1))/t, \tag{11.107}$$

$\lambda_{\max} = 0.37$ for $t = 2$. The number of new captures depends ever more on the behavior of $g(\lambda)$ near $\lambda = 0$, where it cannot be well-estimated, and there is no particular reason to trust Fisher’s negative binomial assumption (11.102).

Table 11.6 *Shakespeare word count data; 14 376 distinct words appeared just once each in the canon, 4343 twice each, 305 eleven times each, etc.*

	1	2	3	4	5	6	7	8	9	10
0+	14 376	4343	2292	1463	1043	837	638	519	430	364
10+	305	259	242	223	187	181	179	130	127	128
20+	104	105	99	112	93	74	83	76	72	63
30+	73	47	56	59	53	45	34	49	45	52
40+	49	41	30	35	37	21	41	30	28	19
50+	25	19	28	27	31	19	19	22	23	14
60+	30	19	21	18	15	10	15	14	11	16
70+	13	12	10	16	18	11	8	15	12	7
80+	13	12	11	8	10	11	7	12	9	8
90+	4	7	6	7	10	10	15	7	7	5

The fascinating formula (11.89) has applications outside the biological realm. Table 11.6 gives a statistical view of Shakespeare. Spevack (1968),

a “canon” or authoritative volume of all Shakespeare’s plays and poems, comprised 884 647 total words, among which were 31 534 distinct words. Of these, 14 376 were so rare they showed up just once each, 4343 twice each, etc. The table gives n_j for $j = 1, 2, \dots, 100$, $n_1 = 14\,376$, $n_2 = 4343, \dots, n_{100} = 5$.

Suppose one found proportion t of a “new” canon, that is, previously unknown Shakespeare works of $884\,647 \cdot t$ total words. How many previously unused words would be seen? The predicted number is given by $\hat{\mu}(t)$ (11.89). For $t = 1$, that is, for a whole new canon of Shakespeare, the prediction⁸ is

$$\hat{\mu}(1) = 11,430 \pm 178. \quad (11.108)$$

Fisher’s model gives $\hat{\mu}(1) = 11\,483$, nearly the same result using MLE estimates $\hat{\alpha} = -0.3954$, $\hat{\beta} = 104.26$.

A variation on missing species is the “missing mass problem.” Define

$$v_0 = \sum_0 \lambda_s / \sum \lambda_s \quad (11.109)$$

where $\sum \lambda_s$ is the sum over all species of the Poisson intensity parameters (11.82), while $\sum_0 \lambda_s$ is the corresponding sum over species *not* seen in the original observation period; v_0 is the missing proportion of species, weighted according to their natural prevalence.

The expected value of the numerator in (11.109) is

$$\sum_s \lambda_s e^{-\lambda_s} = S \int_0^\infty \lambda e^{-\lambda} dG(\lambda) = \eta_1 \quad (11.110)$$

while that for the denominator is

$$\sum_s \lambda_s = \sum_s E\{x_s\} = E\left\{\sum_s x_s\right\} = E\{T\} \quad (11.111)$$

where T is the total number of individuals observed, $T = \sum_j j n_j$. The obvious missing mass estimator is then

$$\hat{v}_0 = n_1/T. \quad (11.112)$$

For the Shakespeare word counts,

$$\hat{v}_0 = 14\,376/884\,647 = 0.016, \quad (11.113)$$

⁸ Checks have been run, with reassuring results, by removing one of the plays, recomputing the now reduced counts n_j , and seeing how well (11.89) predicts the “new” words in the removed material.

suggesting that we have seen all but 1.6% of Shakespeare's vocabulary, when weighted by his usage frequency. (But not nearly that much in a missing species sense: there is at least proportion $11\,430/31\,534 = 0.36$ *distinct* missing words (11.108), and other methods raise this to more than 1.00.)

The early empirical Bayes literature is filled with other fascinating prediction results concerning the hierarchical Poisson model

$$\lambda \sim g(\cdot) \quad \text{and} \quad x|\lambda \sim \text{Poi}(\lambda). \quad (11.114)$$

For example, (11.84) gives, almost immediately, Robbins' prediction formula

$$E\{\lambda|x = j\} = (j + 1)\eta_{j+1}/\eta_j \quad (11.115)$$

and the useful estimate $\hat{E}\{\lambda|x = j\} = (j + 1)n_{j+1}/n_j$, an analog of the normal-theory result (11.28).

Exercise 11.13 Verify (11.115).

The missing species formula, like the James–Stein results of Chapter 1, had a near-magical quality when it first appeared in the 1950s and 60s. But, viewed in the context of this book, these are recognizable, if flamboyant, relatives of our more workaday empirical Bayes methods for estimation, testing, and prediction.

My own belief is that empirical Bayes theory is still in its adolescence. The use of indirect evidence — what I called learning from the experience of others — was not much needed for most of 20th century statistics, and actively disparaged in important areas such as drug testing and clinical trials. A total reliance on direct evidence is an unaffordable luxury in large-scale applications. I expect, or at least hope, that 21st century statisticians will perfect an empirical Bayes theory as complete and satisfying for large-scale problems as linear regression and ANOVA are within their ambit.

Notes

The pamr prediction procedure was developed, under the name “nearest shrunk centroids,” in Tibshirani et al. (2002). Tibshirani and Tibshirani (2009) discuss a bias-correction for the kind of minimum-seeking cross-validation suggested in Table 11.1. A full review of modern prediction methods is available in Hastie et al.'s very readable 2009 text on “statistical learning.” The Ebay empirical Bayes prediction algorithm appears in

Efron (2009b). Subramanian et al. (2005) discuss the Michigan lung cancer study.

The normal hierarchical model (11.22) goes far back into the empirical Bayes literature, and played important roles in the profound investigations of multidimensional normal estimation of Brown (1971) and Stein (1981). Robbins (1956) includes the formula for $E\{\delta|z\}$ in (11.40), prediction formula (11.115), and much more.

Benjamini and Yekutieli (2005) pioneered large-scale effect size estimation, proposing the $BY(q)$ algorithm. A version of Figure 11.5 appears in Efron (2008a).

The missing species problem has a distinguished pedigree, with key contributions by Good (1953), Good and Toulmin (1956) (crediting Alan Turing of Turing machine fame), as well as Robbins (1968). Fisher's original work with Corbet appears in Fisher et al. (1943). The Shakespeare data and analysis is from Efron and Thisted (1976), where linear programming methods are invoked to provide a lower bound of 35 000 on Shakespeare's unseen vocabulary. A later paper, Thisted and Efron (1987), applies versions of formula (11.89) to the question of authenticity of a newly discovered poem attributed to Shakespeare.

Hierarchical models such as (2.47) and (11.114) are available for all one-parameter exponential families, as discussed in Muralidharan (2009).

